Computer Architecture (Practical Class)
Pointers in C

Luís Nogueira Nuno Pereira Raquel Faria

Departamento de Engenharia Informática
Instituto Superior de Engenharia do Porto

{lmn,nap,arf}@isep.ipp.pt

2018/2019

Introduction

- A program sees memory as one large array containing bytes
- We do not use the term "index" when refering to a memory location. We use the term *address*
- When you declare a variable, you are reserving one or more continuous bytes of memory
- Each declared variable has an *address*, which indicates where the variable data starts in the memory
- In IA-32 addresses are 32 bits long, thus can vary between 0 and 4GB

Pointers

- C has special variables, called *pointers*, that are used to store memory addresses
- Pointers are declared like normal variables, with a type associated to it (we will see how this is used later)

Pointers always have the same size

- The size of an address of the underlying architecture (32 bits = 4 bytes in IA32)

- Pointers allow direct access to memory, making it possible to change the values in the memory addresses stored in the pointers
- Some tasks are easier/more efficient to implement when using pointers, and some (such as dynamic memory allocation) are only possible using pointers

Pointers: Use and notation

- Just like any variable in C, a pointer must be declared before being used
- A pointer is declared using `type *` before the variable identifier:

```
int *ptr1; // declares a pointer to an integer
char *ptr2; // declares a pointer to a char
```

- To obtain the address of a variable, use '&' before its identifier:

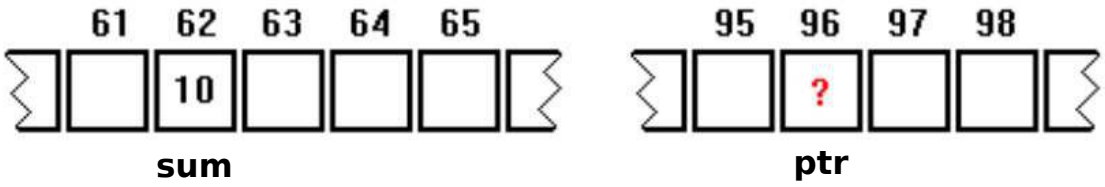
```
int x;
char c;
ptr1 = &x; // store the address of x in ptr1
ptr2 = &c; // store the address of c in ptr2
```

- The *dereference operator* '*' accesses the value at a memory address:

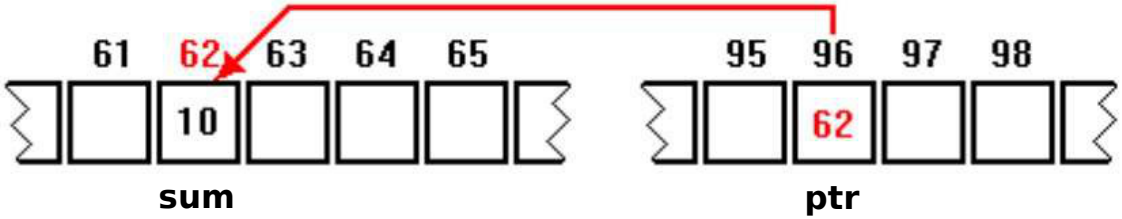
```
*ptr1 = 10; // assign 10 to the value pointed by prt1
*ptr2 = 'X'; // assign 'X' to the value pointed by prt2
```

Content of a variable and its address (1/2)

```
char sum;
char *ptr;
sum = 10;
```

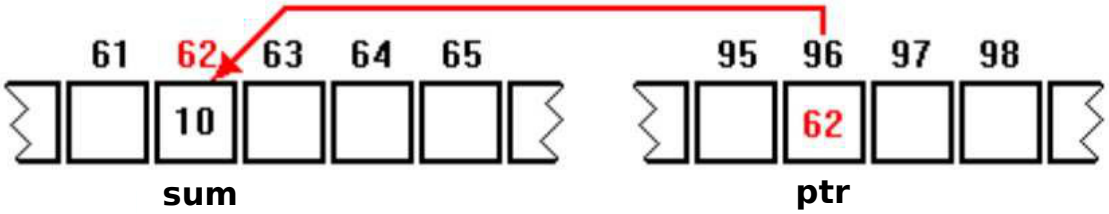



```
ptr = &sum;
```

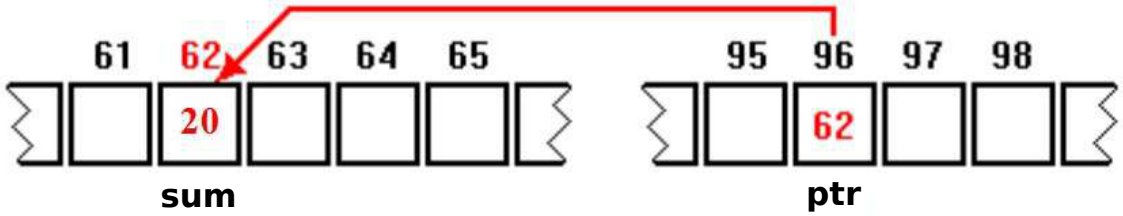


Content of a variable and its address (2/2)

- To access the contents of a memory address in a pointer, use the * operator:




```
*ptr = 20;
```



Pointers should always be initialized to a valid address before being used

- Using an uninitialized pointer has undefined behaviour!

- Wrong way[1]:

```
int x;
int *ptr; // uninitialized pointer

*ptr=22: // this is a very bad idea!
```

- Correct way:

```
int x;
int *ptr;

ptr=&x; // now is initialized to the address of x
*ptr=22: // assign 22 to the memory addressed by ptr (x)
```

[1] This code will, most likely, result in a segmentation fault. A segmentation fault occurs when a program tries to access an invalid memory address. The operating system detects this and terminates the program.

Big/Little Endian

- We have said that the memory is a large array of bytes... So, how do we store data larger than 1 byte?
- Easy: we divide the data into bytes and store it! But, this means we have two ways of storing data in memory:
  - Big Endian
    - Store the most significant byte in the smallest address.
    - Adopted in platforms by Sun, PPC Mac, transferring data on the Internet
  - Little Endian
    - Store the least significant byte in the smallest address.
    - Adopted by x86, ARM

Little Endian example

- Let us assume we want to store the number 305419896 (0x12345678):

31 30 29 28 27 26 25 24	23 22 21 20 19 18 17 16	15 14 13 12 11 10 9 8	7 6 5 4 3 2 1 0
0x12	0x34	0x56	0x78

- Byte by byte

7 6 5 4 3 2 1 0	
0x78	} Address x
0x56	} Address $x + 1$
0x34	} Address $x + 2$
0x12	} Address $x + 3$

Big Endian example

- Let us assume we want to store the number 305419896 (0x12345678):

31 30 29 28 27 26 25 24	23 22 21 20 19 18 17 16	15 14 13 12 11 10 9 8	7 6 5 4 3 2 1 0
0x12	0x34	0x56	0x78

- Byte by byte

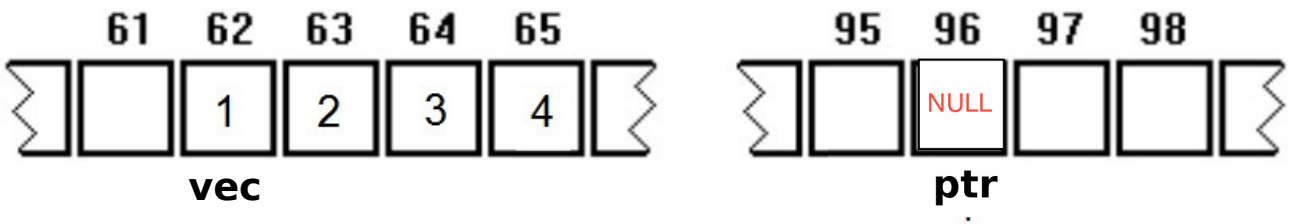
7 6 5 4 3 2 1 0	
0x12	} Address x
0x34	} Address $x + 1$
0x56	} Address $x + 2$
0x78	} Address $x + 3$

Listing 1: data-rep.c (http://codepad.org/leCxR8yX)

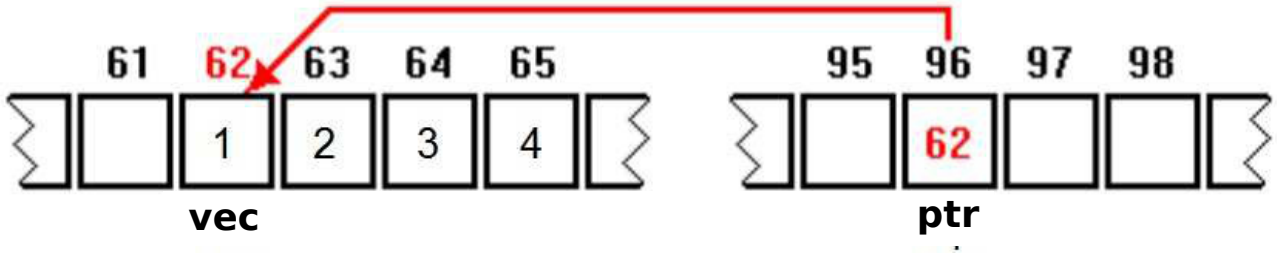
```
#include<stdio.h>
int main(){
    unsigned int i = 0xFFAAEEBB;
    int b = 0x1234;
    char c = 'A';
    unsigned int x = 0x12345678;
    // %p: Pointer address format specifier
    printf("i em %p\n",&i);
    printf("b em %p\n",&b);
    printf("c em %p\n",&c);
    printf("x em %p\n",&x);
    return 0;
}
```

Variable	**Address**	**Memory** (byte by byte in hexadecimal)
i	0xfff091c8	bb ee aa ff
b	0xfff091c4	34 12 00 00
c	0xfff091c0	00 00 00 41
x	0xfff091bc	78 56 34 12

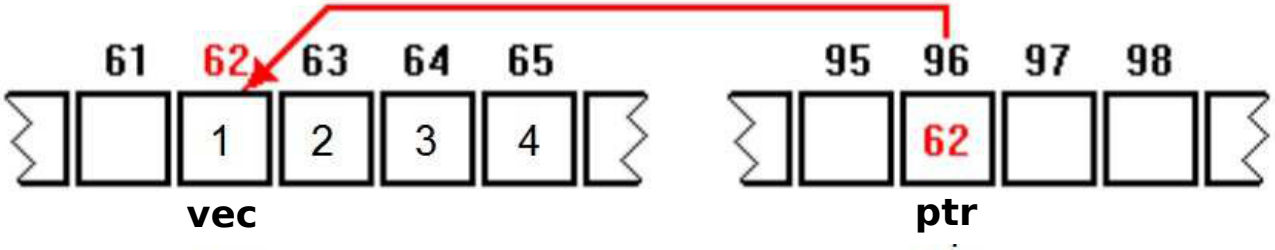
```
char vec[4] = {1,2,3,4};
char *ptr = NULL;
```




```
ptr = vec;  /* in the case of arrays, the identifier is the address;
               so, no need for '&' */
```



Pointer arithmetic (1/4)



- You can do *pointer arithmetic*:

```
ptr++; // ptr now is a pointer to the second element of vec
printf("%d", *ptr); // will print "2"

ptr+=2; // ptr now is a pointer to the fourth element of vec
printf("%d", *ptr); // will print "4"
```

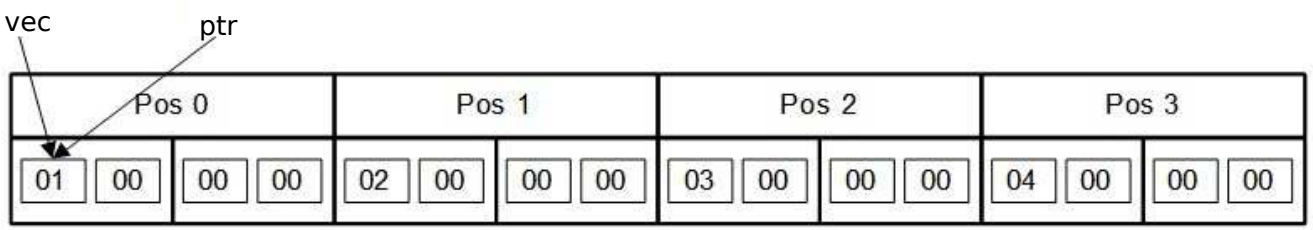
Pointer arithmetic (2/4)

- The type of the pointer is relevant when performing pointer arithmetic (the compiler decides on the number of bytes added or subtracted based on the pointer type)

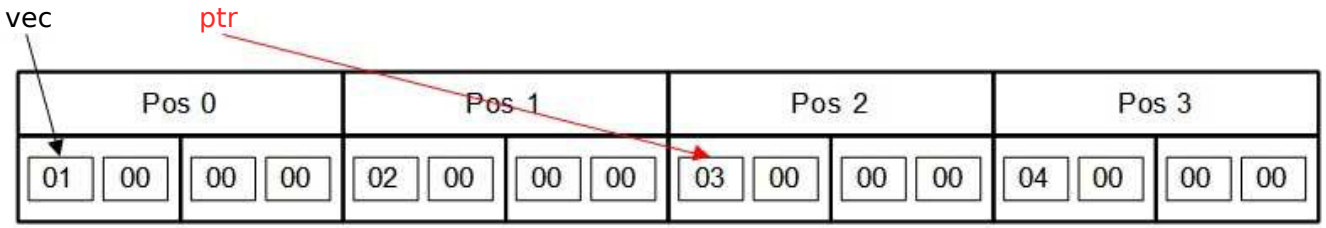
```
int *ptr_total; /*(4 Bytes increments/decrements in IA32)*/
char *ptr_name; /*(1 Byte increments/decrements in IA32)*/
```

- Example:

```
int vec[4] = {1,2,3,4};
int *ptr = vec;
```



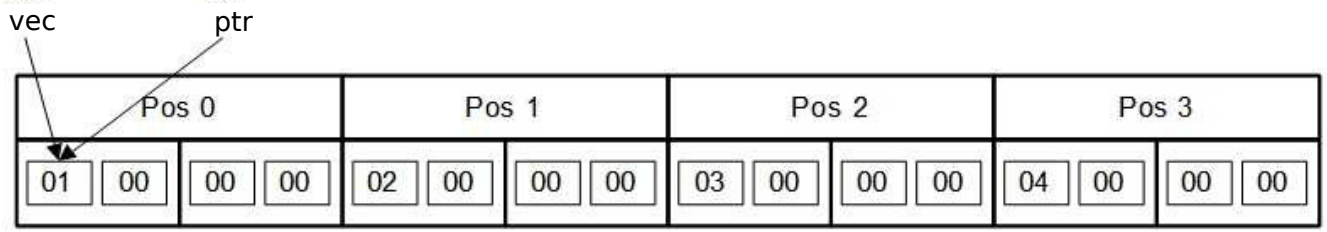

```
ptr = ptr+2;
```



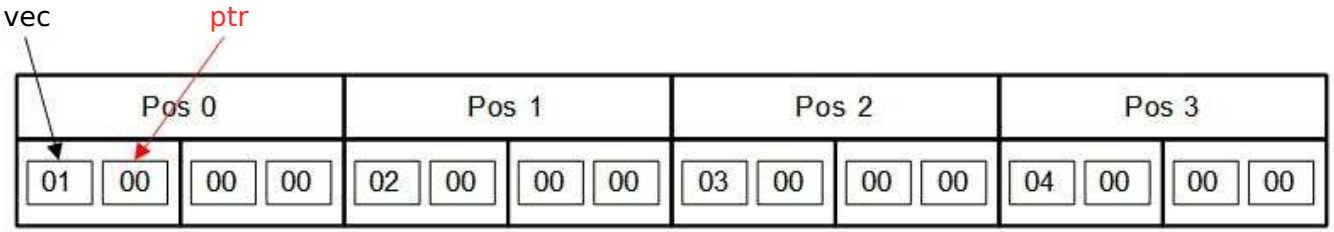
Pointer arithmetic (3/4)

- Now we make `ptr` a `char*`...

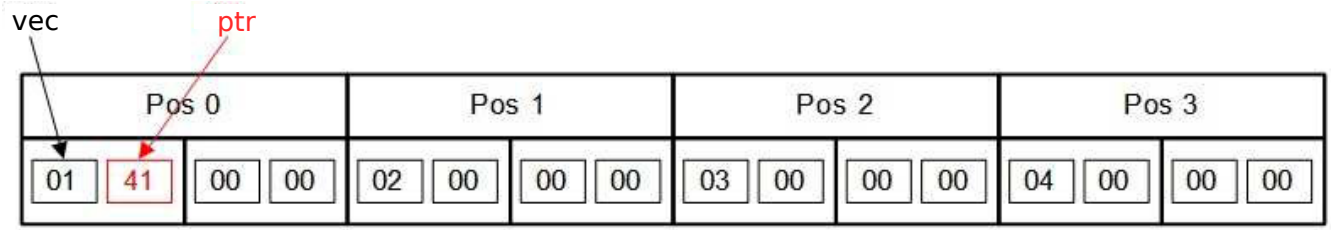
```
int vec[4] = {1,2,3,4};
char *ptr = (char *)vec; // a cast is needed to avoid a warning...
```




```
ptr = ptr+1;
```




```
*ptr = 'A';
```



Pointer arithmetic (4/4)

- Active learning activity: What is the output of the following code ?
  - A. 1 2 3 4 5
  - B. 2 3 4 5 6
  - C. 1 3 5 7 9
  - D. None of the above.

Listing 2: pointer-arith.c (http://codepad.org/V3qmKQkp)

```
#include<stdio.h>

int main(){
        char arr[]={1,2,3,4,5,6,7,8,9,10};
        short *ptr=(short*)arr; // cast is needed to avoid a warning...
        int i;
        for (i=0; i<5; i++) {
                arr[i] = *( ptr + i );
                printf("%d ", arr[i]);
        }
        return EXIT_SUCCESS;
}
```

- Print data to the console using printf and puts

Listing 3: output.c (http://codepad.org/swgi0sNG)

```
#include<stdio.h>

int main()
{
        int a = 2, b = 3;
        int *c = &a, *d = &b;
        printf("Value: %d\n", *c); /*prints the value 2*/

        printf("Address: %p (%p)\n", &a, c);
        /* outputs the address of a*/

        if (*c == *d) puts("Same value"); // false
        *c = 3;
        if (*c == *d) puts("Now same value"); // true
        c = d;
        if (c == d) puts ("Now same address"); // true
        return EXIT_SUCCESS;
}
```

- Reading from the console can prove problematic... here are some solutions.

Listing 4: input.c (http://codepad.org/nmuqfS91)

```
#include<stdio.h>
#include <stdlib.h>
int main()
{
        int n1=-1, n2=-1, n3=-1;
        int *ptr = &n2;
        char buf[BUFSIZ], c;
        /* read integers using scanf(); pass the address of the var. */
        printf ("Enter a number: ");
        scanf("%d", &n1);
        printf ("Enter another number: ");
        scanf("%[^0-9]%d", buf, ptr); // garbage goes to buf...
        /* You will notice that scanf is a bit problematic...
           An option, is to read a string and convert as needed
           First, flush characters scanf() leaves in the input */
        while((c = getchar()) != '\n' && c != EOF) /* discard */;
        printf ("Enter yet another number: ");
        if (fgets(buf, sizeof(buf), stdin) != NULL) {
                n3 = atoi(buf); // returns 0 if conversion fails
                printf ("You entered %d %d %d\n", n1, n2, n3);
                return EXIT_SUCCESS;
        }
        return EXIT_FAILURE; /* if reaches here, fgets() failed */
}
```

Practice

1. Write the representation in memory, in Big endian and Little endian of the following values:
   1. 0x1188 (16 bits)
   2. 0xff3455b6 (32 bits)
   3. 0x28934def (32 bits)
2. Correct the following code:

Listing 5: exerc01.c

```
int main ()
{
  int * ptr ;
  int i;
  int soma=0;

  for (i=0; i<10; i++)
       {
        scanf("%d", ptr);
        soma = soma + *ptr;
        }
  printf("Soma= %d \n", soma);
  return 0;
}
```